Homework — 1.2.3 Software Development

Name: ________________________ Date: ___________ Mark: ______ / 20

OCR H446 — set after the 1.2.3 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. The five main stages of the **waterfall** lifecycle are shown below in a jumbled order. Number them **1–5** to show the correct order. **[2]** *(AO1)*

Stage	Order (1–5)
Testing	
Analysis (requirements)	
Evaluation / maintenance	
Design	
Development (coding)	

2. Tick **True** or **False** for each statement. If a statement is false, write a correction in the final column. **[3]** *(AO1)*

Statement	True / False	Correction if false
In extreme programming, developers always write code alone so that they can concentrate.		
In the waterfall model, the client is mainly involved at the start (requirements) and the end (evaluation), not during every stage.		
There is no point testing a program with erroneous data, because the program is supposed to reject it anyway.		

3. A program accepts a percentage mark that must be between **0 and 100 inclusive**. Complete the table of test data for this input. **[3]** *(AO2)*

Type of test data	Example value	What this test checks
Normal (valid)	(a)	(b)
Boundary	(c)	(d)
Erroneous	(e)	(f)

4. Match each methodology to its description by writing the correct **letter** in the Match column. Two descriptions are not needed. **[2]** *(AO1)*

Methodology	Match
Waterfall	
Extreme programming	
Rapid application development	
Spiral	

Letter	Description
A	A prototype is built quickly, then repeatedly refined using user feedback until it becomes the final system
B	Development passes through a sequence of stages, each completed and signed off before the next begins
C	Each loop starts by setting objectives and analysing risks before building and testing an increment — suited to large, risk-heavy projects
D	Emphasises code quality through practices such as pair programming, test-first development and frequent small releases
E	The whole program is translated into machine code in one go before it is run
F	Modules are tested individually using knowledge of the code inside them

5. A company is building software for a client whose requirements are **likely to change frequently** during the project.

(a) Explain why an **agile** methodology is more suitable than the **waterfall** model for this project. **[3]** *(AO2)*

(b) The company considers using **rapid application development (RAD)** instead. State **one** drawback of RAD. **[1]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

6. State **two** differences between a **compiler** and an **interpreter**. **[2]** *(1.2.2)*

7. A **washing machine** and a **server cluster** use different types of operating system. Name the most suitable OS type for each and justify your choice. **[2]** *(1.2.1)*

8. Explain the difference between **RAM** and **ROM** in terms of volatility and typical use. **[2]** *(1.1.3)*